

Instantaneous emotion sharing: Early brain signatures of understanding expressive communicative actions

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ABSTRACT

Words and sentences can be used to carry out a wide range of different communicative actions. Recognizing these actions quickly is crucial for effective understanding. Previous research indicates that communicative action understanding is a rapid process, as brain signatures of different communicative actions differ within ca. 250 ms following the relevant linguistic information. However, previous work has explored a restricted set of communicative actions, leaving other ones still unexplored. In the present study, we examine emotion-related utterances in the 1st person singular ("I am happy"), which are typically applied to express an emotional state of the speaker, that is, to perform so-called a 'expressive' communicative action. We recorded brain potentials while subjects read such 'expressive sentences' and similar ones in the 3rd person singular, which are likely applied to assert or report facts ("he is happy"). Further assertive control past-tense sentences were included to control for any 1st/3rd person differences. Brain responses to sentences with expressive communicative actions differed from those to assertives 221–270 ms after the critical sentence-final word appeared, and an even earlier dissociation was observed. These results extend previous findings from contrasting assertive and directive communicative actions by demonstrating that the brain is quick in discriminating between communicative actions, therefore suggesting a generalizable temporal pattern. Since communicative action is processed fast, within ca. 250 ms after the critical utterance-related information is presented, our results are consistent with parallel models of language processing stating that pragmatic processing occurs rapidly and near-simultaneously with that of semantic and morphosyntactic information.

1. Introduction

In our daily life, we seamlessly use language to perform communicative actions in the context of actions of our interlocutors. Therefore, communicating with others is a form of joint action (Clark, 1996), whereby dialogue is the prototypical scenario of linguistic interaction (Clark, 1996; Fritz and Hundsniurser, 1994; Kuhlén and Abdel Rahman, 2023; S. Levinson, 2012; Pickering and Garrod, 2004; Wittgenstein, 1953). In such a framework, language is used alternately by the interlocutors so that the roles of the speaker performing linguistic actions and the listener understanding them cannot easily be separated, in particular because linguistic actions are influenced online by the other interlocutors' immediate responses and signals (both verbal and non-verbal). Language allows us to carry out a sizable variety of different communicative actions, including *stating*, *requesting*, *thanking*,

accusing, etc. (Austin, 1962; Searle, 1969, 1979; Wittgenstein, 1953). However, conversation is tightly timed, whereby a new turn normally starts about 200 ms after the end of the previous one (Levinson, 2016; Meyer, 2023; Stivers et al., 2009), or even overlaps in time with it. Yet, to respond appropriately to a verbal communicative action, one does not only need to understand the semantic content of a verbal utterance but also the type of communicative action performed by it. Therefore, recognizing the pragmatic function of an utterance used by the speaker is a further prerequisite of successful communication (Levinson, 2012). From a classical linguistic-pragmatic perspective, these actions are typically referred to as *speech acts* with specific *illocutionary forces* (Austin, 1962; Searle, 1969). Henceforth, we use the term *communicative action* as a synonym for *communicative function*, *social action* (Holler, 2025), *illocutionary force* and (loosely speaking) *speech act*. Thereby, we emphasize the intentional communicative nature of such actions, but

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also that these actions can be conveyed by non-linguistic means (Holler, 2025). In the present work, we address the broader question of the time course of neural processing of communicative actions. Among the variety of possible communicative activities, we focus on expressive communicative actions, which express a psychological state of the speaker (Searle, 1979), such as their own emotions or attitudes (Searle, 1979; Guiraud et al., 2011; Norrick, 1978). Classic examples of expressives are *congratulating, thanking, or expressing regret or joy*. The need to share psychological states has been suggested to be among the three fundamental functions of human communication together with requesting (directives) and informing (assertives) (M. Tomasello, 2008), with the latter two having received more attention in neurolinguistic research in the past years (R. Tomasello, 2023; R. Tomasello et al., 2025). Expressives differ from other communicative actions in that they express the speaker's emotional state, carrying distinct social, emotional and affective consequences. This makes them theoretically significant, as their processing may depend more extensively on socio-emotional neural systems and self-related perspective-taking mechanisms (e.g., Barrett, 2017; LeDoux and Brown, 2017; Lindquist et al., 2012) than assertives or directives do, thus possibly engaging different neural processes and resulting in different temporal dynamics.

Critically, not any sentence containing emotional words (e.g., “happy”) can be easily used to carry out an expressive communicative action. For instance, a sentence in the second or third person may contain such words (“he is happy”) and may attribute the corresponding emotional state to a person referred to as “he”, but such utterances usually are not used to express and share the inner states and emotions of the speaker themselves. In contrast, a first-person sentence such as “I am happy” is more appropriate to this purpose and is more likely to be used and understood as a tool to express the speaker's inner state. Therefore, containing emotional words is neither sufficient nor necessary for a sentence to be an appropriate tool for performing an expressive communicative action. Therefore, one explicit and paradigmatic way of performing expressive actions is by way of uttering ‘internal-state sentences’ in the 1st person singular, for example, “I am happy” (Searle, 1979). Interestingly, a minor syntactic variation of such sentences, for example, by transforming them into the 3rd person singular (“S/He is happy”), results in an utterance that is invalid as a tool for performing expressives. These sentences can be used for describing the feelings or ‘internal states’ of others, but not for expressing them. A minor grammatical change in a sentence can, therefore, result in a fundamental change in communicative action: from an expressive to an assertive communicative action (Searle, 1979).

We note that previous studies have examined the neural basis of expressive communicative actions that convey *attitudes* rather than emotions. For instance, single-word swears/interjections such as “damn” or “shit” (Donahoo and Lai, 2020) were initially processed similarly to negatively valenced words; their ERPs started to diverge only around 550 ms. Similar expressive expressions (“The damn dog”) compared to descriptive ones (“The black dog”) were found to alter the neurophysiological responses to the referring expression (i.e. “dog”) early on, starting at 150–250 ms after noun onset. Still, no ERP differences were found for the adjective itself (Donahoo et al., 2022). Similarly, videos of people saying person-descriptive words (e.g. “loser”, “brave”) vs. non-person-descriptive ones (“timeless”, “tragic”) to the experimental participant were found to elicit different brain responses between 400 and 700 ms after critical word onset (Rohr and Abdel Rahman, 2018). Whereas the latter study was not explicitly framed to investigate expressive communicative actions, the experimental subjects might have interpreted the person-descriptive condition as including an expressive communicative action. In sum, previous work on these attitude-expressives overall pointed to a rather late processing of this communicative action, from 400 ms after critical stimulus onset (Donahoo et al., 2022; Donahoo and Lai, 2020; Rohr and Abdel Rahman, 2018). To the best of our knowledge, the early phase of processing of emotion-expressives remains unexplored.

Expressive communicative actions (expressing emotions or attitudes) are only some of the many actions that can occur in communication. Several neuropragmatic studies have been conducted on other communicative actions, in particular on assertive and directive ones (Antoine et al., 2024; Boux et al., 2021; Chang and Wang, 2022; Egorova et al., 2013, 2014, 2016; see R. Tomasello 2023; Tomasello et al., 2019, 2022, R. Tomasello et al. 2025 for a review) as well as a few other communicative actions such as offer declinations and pre-offers (Gisladdottir et al., 2015, 2018) as well as promises (Chang and Wang, 2022). Earlier studies therefore focused on the time course of communicative action processing, finding that brain responses to naming and request communicative actions differed already early on within the first 100–300 ms post-critical stimulus onset and therefore well before the N400 brain response, which earlier work had associated with semantic (Kutas and Federmeier, 2011) and pragmatic processing (for a review see Canal and Bambini, 2023). Such early neurophysiological distinctions of communicative actions were found both when the communicative action was predictable by the preceding context (Antoine et al., 2024; Boux et al., 2021; Egorova et al., 2013, 2014, 2016; Gisladdottir et al., 2015, 2018) and when it was not predictable (R. Tomasello et al., 2019, 2022). Studies based on high-spatial resolution methods such as fMRI, MEG and EEG source reconstruction indicated that different brain systems were active for these two different types of communicative action. Requests were found to activate the motor system more than naming communicative actions both in comprehension (Egorova et al., 2014, 2016; R. Tomasello et al., 2019, 2022) and production (Boux et al., 2021). Overall, these findings were robust across a variety of cues that were used to mark communicative action: from previous linguistic context (Egorova et al., 2013, 2014, 2016), to situational context (Boux et al., 2021) prosody (R. Tomasello et al., 2022) and simultaneous word-gesture combinations (R. Tomasello et al., 2019). Although investigating assertives and directives has been ground-laying for our understanding of how communicative actions are processed by the brain, many different types of communicative actions are expressed in dialogues. In light of this diversity, it becomes crucial to investigate brain activity for hitherto unexplored types of communicative actions, which is essential to understand whether previous findings can be generalized.

Here, we investigated the neurophysiological basis of the comprehension of expressive communicative actions performed with the intention to share one's own emotion. We ask whether specific brain signatures of expressives can be identified. For this purpose, sentences that can be used for expressive actions (e.g., “I am happy”) were contrasted with sentences that are not well-suited for it (e.g., “He is happy”) and which, due to their third person form, are instead more appropriate for performing assertive actions. Because a change in grammatical person is crucial for this change in communicative action, further control conditions capturing the mere effect of formal grammatical person change in the absence of any communicative action change, were added (e.g., “I am blond” vs. “He is blond”, which can both function as assertives but not as expressives). Therefore, we implemented a 2 Person $\times$ 2 Adjective design where only the sentences containing an emotional (vs. physical) adjective in the 1st person (vs. the 3rd person) could function as a genuine expressive, whereas all other conditions functioned primarily as assertive control conditions. Based on previous findings for non-expressive communicative actions (e.g., Egorova et al., 2013, 2014, R. Tomasello et al., 2019, 2022), we hypothesized that early dissociations between expressives and assertives should be detected at the neural level. In addition, we expected these dissociations to be manifest early (within ca. 250 ms) after presentation of the critical stimulus (the word completing the sentence/utterance) (Pulvermüller, 2018). The presumed brain signature of communicative function was predicted to become manifest in a significant interaction effect between grammatical person and adjective type, whereby a neurophysiological difference was expected between adjective types in the 1st person sentences, but not between those in the 3rd person.

2. Material and methods

2.1. Experimental subjects

28 subjects took part in this study (15 females, 13 males). Our sample size was determined based on previous EEG studies investigating the neural basis of communicative action processing (Boux et al., 2021; Donahoe et al., 2022; Egorova et al., 2013; R. Tomasello et al., 2019) as well as other linguistic-pragmatic phenomena (e.g., Bambini et al., 2019; Masia et al., 2017). The a priori inclusion criteria for this study were: (i) being between 18 and 35 years old; (ii) being psychologically and neurologically healthy; (iii) being a German native speaker who grew up in a monolingual environment; (iv) being right-handed. Subjects reported no reading/writing disorders and had normal or corrected-to-normal vision. Three subjects were excluded due to technical problems during data collection, whereas for two additional subjects, only the first two blocks could be included in the final analyses. One subject was excluded because of poor performance (i.e., $d' = 0$; see Results Section 3.1) in the control memory task (see Section 3.1). Therefore, the final sample entering the analysis consisted of 24 subjects. They were on average 24.3 ± 4.2 SD years old and were confirmed to be right-handed, with an average IQ of 83.5 ± 19.1 SD as assessed by the Edinburgh Handedness test (Oldfield, 1971). All study procedures were approved by the local Ethics Committee of the Charité University Hospital (approval number: EA4/138/19). Participants provided written informed consent and were compensated for their participation with 10 EUR/hour.

2.2. Stimulus material

Our stimulus material consisted of three-word sentences in German (see Table 1). Syntactically, their structure was identical. The first word was a singular personal pronoun (1st vs. 3rd person), the second was the verb “be” accordingly conjugated, and the third (critical) word was an adjective (emotional vs. physical). Depending on the combination between the grammatical person and the type of adjective, four different conditions were possible: 1st/emotional (“*ich bin froh*” meaning “I am happy”), 1st/physical (“*ich bin blond*” meaning “I am blond”), 3rd/emotional (“*er ist froh*” meaning “he is happy”) and 3rd/physical (“*er ist blond*” meaning “he is blond”). Of these four conditions in the 2 Person $\times$ 2 Adjective design, only the 1st/emotional could function as a genuine expressive, whereas the other three were primarily compatible with an assertive.

All emotional and physical adjectives could be appropriately and unambiguously used in describing humans. Emotional adjectives described a person's emotional state (e.g., happy, bored, amused, etc.) and did not include evaluations about a person (e.g., boring, funny). Physical adjectives referred to general body characteristics (e.g., tall, slim, chunky, etc.) and also did not include evaluative adjectives (e.g., ugly, beautiful). Emotional and physical adjectives were matched for various psycholinguistic properties (see Table 2). Independent t-tests confirmed that they were matched for length measured both in letters (p

Table 2

Summary of the psycholinguistic properties of emotional and physical adjectives. Normalized lemma, bigram and trigram frequencies were taken from the dlexDB database (Heister et al., 2011).

	emotional		physical		p-value
	mean	SD	mean	SD	
Length (letters)	7.82	1.24	7.18	2.36	0.129
Length (syllables)	2.29	0.46	2.04	0.74	0.162
Normalized lemma frequency (log 10 normalized)	0.55	0.56	0.79	0.87	0.357
Average normalized bigram frequency	39525.66	12317.24	38432.00	16075.36	0.327
Average normalized trigram frequency	14145.69	5229.87	17591.91	11086.28	0.276
Number of morphemes	2.62	0.70	1.91	0.75	0.001*
of which affixes	1.59	0.70	0.85	0.74	0.000*
of which derivational affixes	0.91	0.51	0.50	0.51	0.040*
of which inflexional affixes	0.68	0.47	0.32	0.53	0.001*

= 0.129) and syllables (p = 0.162). Furthermore, they were matched for lemma frequency (p = 0.357, after log10 transformation to meet the normality assumption), average normalized bigram (p = 0.327) and trigram frequency (p = 0.276) which were all taken from the dlexDB database (Heister et al., 2011). In addition, the morphological complexity of the emotional and physical adjectives was assessed by counting the average number of affixes (derivational and inflexional separately) and morphemes. A Chi-squared (χ^2) test indicated that emotional adjectives scored higher than physical ones on all morphological properties, indicating that they were overall morphologically more complex (all p < 0.05).

In addition to the psycholinguistic properties of the critical words, we also assessed the properties of the entire utterances, i.e., of the critical words embedded in the 1st and 3rd person sentences. These properties were analyzed with a mixed ANOVA with Person [1st, 3rd] as a within-item factor and Adjective [emotional, physical] as a between-item factor. First, we performed a corpus analysis estimating the conditional probabilities of each adjective following either the 1st or 3rd singular person pronouns in the German Reference Corpus (Deutsches Referenzkorpus, DeReKo-2020-I, W – Archiv der geschriebenen Sprache, W-öffentlich, <https://cosmas2.ids-mannheim.de/cosmas2-web>) which included German written language from literary, scientific and scientific divulgation texts as well as newspapers. This was done by searching for the absolute frequency of each adjective following (immediately or not) either the 1st or 3rd person pronoun within a sentence. These values were then divided by the absolute frequency of the pronoun. To meet the assumption of Gaussian distribution, these relative frequencies underwent log normalization before statistical testing. The relative frequencies (see Table 3) were not explained by a main effect of Person, Adjective, or by an interaction between the two (p > 0.05) and were therefore comparable across groups. Next, valence ratings, arousal ratings and communicative action types associated with the stimulus sentences were provided by an independent sample of 12 participants. Valence and arousal were assessed on a 0-100 analog scale. Furthermore, subjects were asked to categorize the different sentences as expressive or assertive communicative actions in a two-alternative forced choice task. Subjects were asked which option best describes the intention of the speaker. They were instructed to click on “express

Table 1

Example of the stimulus material in all four experimental conditions. 34 different adjectives (underlined) were used for each the emotional and physical conditions and acted as critical words.

Person	Adjective	Example	N items	Communicative Action
1st	emotional	<i>ich bin <u>froh</u></i> (I am <u>happy</u>)	34	expressive
3rd	emotional	<i>er ist <u>froh</u></i> (he is <u>happy</u>)	34	assertive
1st	physical	<i>ich bin <u>blond</u></i> (I am <u>blond</u>)	34	assertive
3rd	physical	<i>er ist <u>blond</u></i> (he is <u>blond</u>)	34	assertive

Table 3

Mean values and standard deviations (SD) of the normalized conditional frequency of the stimulus adjectives as a function of Adjective (emotional vs. physical) and Person (1st vs. 3rd).

	emotional				physical			
	1st person		3rd person		1st person		3rd person	
	mean	SD	mean	SD	mean	SD	mean	SD
Normalized conditional frequency	-4.39	0.64	-4.45	0.53	-4.60	0.67	-4.59	0.66

internal state" if they thought that a speaker uttering such a sentence wants to express their own inner state or feelings. Conversely, they were instructed to select the "describe state of affairs" if the speaker uttering such a sentence describes a state of affairs with their utterance and reports facts.

The analysis of valence (Fig. 1A) indicated a main effect of Person ($F(1,66) = 22.11$, $p < 0.001$), so that sentences in the 1st person had a slightly more positive valence than those in the 3rd person. No main effect of Adjective nor, importantly, an interaction between Adjective and Person was found (both $p > 0.05$). The absence of interaction indicates that our expressive stimuli did not differ in valence from the assertive ones. For arousal (Fig. 1B), a main effect of Person ($F(1,66) = 101.35$, $p < 0.001$) indicated that 1st person sentences were more arousing than 3rd person ones. At the same time, a significant main effect of Adjective ($F(1,66) = 210.31$, $p < 0.001$) indicated that emotional adjectives were more arousing than physical ones. The latter validates our grouping into 'emotional' and 'physical' subsets. Critically, however, a significant interaction between Person and Adjective ($F(1,66) = 26.91$, $p < 0.001$) emerged due to the fact that arousal ratings for 1st/emotional were higher than the additive main effects of Person and Adjective could have explained (all four pairwise comparisons had $p < 0.05$). This would indicate, arguably unsurprisingly, that our expressive stimuli were more arousing than the assertive controls. Finally, the frequency of interpretation of the individual constructions as expressive vs. assertive actions agreed with the classification underlying this experiment (Fig. 1C). The 1st/emotional sentences were classified as expressive actions in 97.79% of cases, whereas the other conditions were rarely classified as expressives (3rd/emotional: 1.72%; 1st/physical: 6.13%; 3rd/physical: 0.49%), but, in the majority of cases, as assertive ones instead. This pattern was explained by a main effect of Adjective ($F(1,66) = 1800.19$, $p < 0.001$) and of Person ($F(1,66) = 2615.47$, $p < 0.001$) and, crucially, by an interaction between Adjective and Person ($F(1,66) = 2067.78$, $p < 0.001$). Post-hoc tests indicated that both physical adjectives ($p < 0.001$) and emotional ones ($p < 0.001$) were affected by Person change, but this effect was modest for the former and pronounced for the latter.

(1,66) = 2067.78, $p < 0.001$). Post-hoc tests indicated that both physical adjectives ($p < 0.001$) and emotional ones ($p < 0.001$) were affected by Person change, but this effect was modest for the former and pronounced for the latter.

2.3. Experimental procedure

The main experimental task was programmed in PsychoPy 3.0.5 (Peirce et al., 2019). Subjects were seated at about 70 cm from a computer monitor onto which visual stimuli were presented. The sentences were presented word by word in a rapid serial word presentation (RSVP) fashion. This presentation modality is known to minimize saccadic eye movements during reading (see e.g., Rubin and Turano, 1992), which are typical artifacts in EEG recording. Individual words and fixation crosses were displayed in lowercase in black Arial font against a grey background. The trial was structured as follows. First, a fixation cross was presented with a duration ranging from 1900 to 2100 ms (in steps of 17 ms, uniformly distributed). This was followed by the presentation of the three words for 150 ms each, interleaved with a fixation cross of 800 ms (see Fig. 2). Subjects were instructed to imagine that the sentences they read were said to them by a close (male) friend in a casual conversation. They were informed that the friend could be talking about himself or about an absent third party, or that he could ask the subjects to do things for him or talk about plans he had with other friends. Because some adjectives might be evaluated differently when attributed to a male or female person (e.g. "petite"), we aimed at keeping the gender of the person they were referring to constant across conditions. We therefore instructed the participants that they should imagine that the sentences they read on the screen were spoken by a male friend so that the 1st person sentences would refer to the male friend himself. In addition, the 3rd person sentences were always formulated in the masculine form ("he" rather than "she") so that they would also clearly

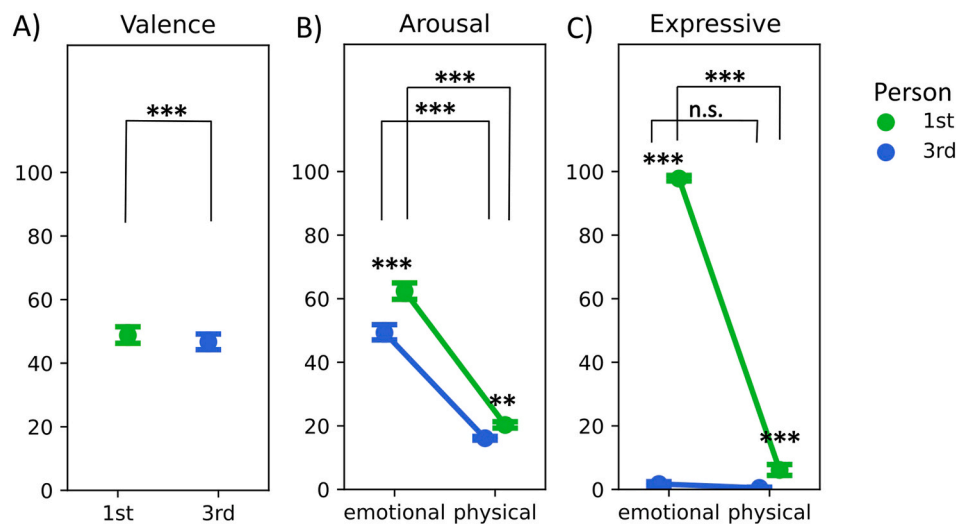


Fig. 1. (A) Main effect of Person on the valence ratings (range: 0-100). (B) Significant interaction of Person and Adjective on ratings of arousal (range: 0-100), with significance asterisks referring to pairwise post-hoc comparisons. (C) Interaction of Person and Adjective on the rate of interpretation as an expressive (%), with significance asterisks referring to pairwise post-hoc comparisons. In all three panels error bars indicate the standard error of the mean and asterisks indicate statistical significance according to the standard notation (* indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$).

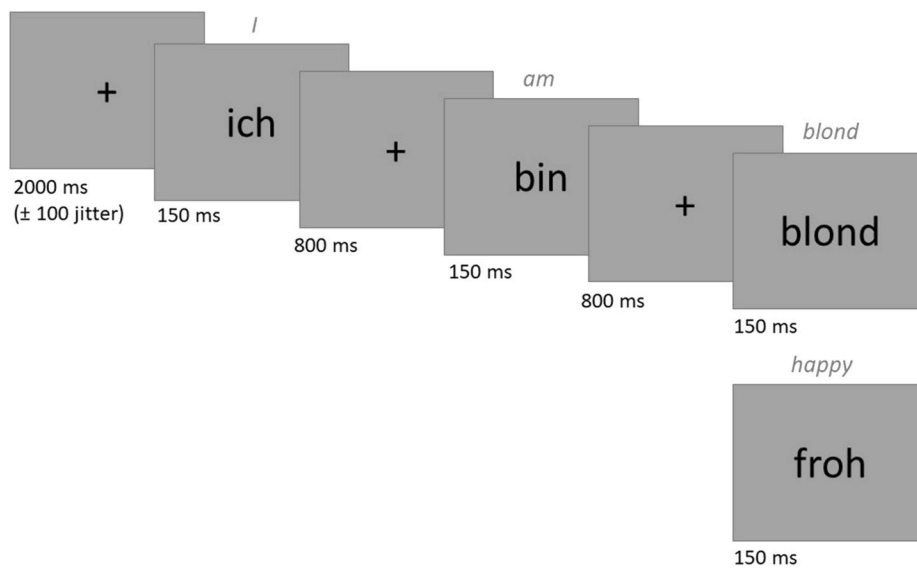


Fig. 2. Illustration of the trial structure in the main passive task exemplified with a stimulus “ich bin blond” (=I am blond) from the 1st/physical condition.

refer to a male person. Overall, the instruction also emphasized that the friend was close, with the intention to make the communicative actions of the task plausible to the experimental subjects, in particular expressives which might be more likely to occur in conversations between close friends. The subjects were instructed to listen carefully and focus on understanding the message conveyed by the friend.

The task was distributed in four blocks each lasting about 10 min, for a total task duration of about 39 min. Two “1st person” blocks each contained 34 1st/emotional and 34 1st/physical trials. Another two “3rd person” blocks each contained 34 3rd/emotional and 34 3rd/physical trials. In addition, all four blocks included filler trials maintaining a three-word sentence structure but that functioned as other types of communicative actions, that is, 34 directive trials (e.g., “gib ihnen Tee” = “give them tea”) and 34 commissive trials (e.g. “ich werde bügeln” = “I will iron”). Data from filler trials was not evaluated. This approach, combined with the use of filler communicative actions (commissives and directives) in both types of blocks, was aimed at maintaining a certain variety of communicative actions and not having a too large difference in probability between the two critical communicative actions. This resulted in a 1st person block where expressives and assertives were equally likely with each a 25% probability and in a 3rd person block where assertives had a 50% probability. If both expressives and assertives had been presented in an intermixed fashion in the same block, then expressives would have had a total probability of 25% making them much more surprising than assertives, which would have had a probability of 75%. The block order was counterbalanced across subjects such that half of the subjects had the order 1st-3rd-1st-3rd, whereas the other half had 3rd-1st-3rd-1st. Each stimulus of interest was therefore repeated twice. The stimuli were repeated to achieve a good signal-to-noise ratio, which is a standard approach in EEG studies to enhance the signal associated with the cognitive processes of interest while minimizing unrelated background activity (Cohen, 2014; Handy, 2004; Luck, 2014). Importantly, all conditions were equally repeated, ensuring that potential repetition-related effects were evenly distributed and could not bias one condition over another.

To assess whether subjects attended the passive task, subjects were invited to complete an unannounced recognition memory task immediately following the EEG recording. The task consisted of a paper-and-pencil form with a list of 40 adjectives: 10 emotional adjectives from the task, 10 novel emotional adjectives, 10 physical adjectives from the task and 10 novel physical adjectives. Subjects were instructed to indicate with a cross whether they remembered seeing each individual adjective in the task.

2.4. EEG recording and preprocessing

The EEG data was recorded while the subjects took part in the passive communicative action processing task while sitting in an electrically and acoustically insulated chamber. A 64-active-electrodes set up (Brain Products GmbH, Munich, Germany) in a 10-10 layout was used. The following modifications were applied: the reference was moved from the FCz position to the tip of the nose and the electrode occupying the PO10 position replaced the empty FCz position. The PO9 and FT9 electrode positions were reassigned as EOG channels placed below and above the left eye, respectively, and the FT10 electrode to the right outer canthus. All electrodes were referenced to the reference electrode placed on the tip of the nose. Data were amplified and recorded using the Brain Vision Recorder (version: 1.20.0003; Brain Products GmbH) and were sampled at 500 Hz. Impedances of the electrodes were kept below 10 K Ω .

The EEG data were processed in MNE Python 1.5.1 (Gramfort et al., 2013), with Python 3.9.13. First, the data were down-sampled to 250 Hz. Next, a band-pass filter at 0.1 – 30 Hz, and a notch filter at 50 Hz were applied, both being finite impulse response filters with a –6dB roll-off. The signal from the upper and lower eye electrodes was used to generate bipolar vertical EOG signals. The horizontal EOG was computed by subtracting the right outer canthus potentials from the average of the upper and lower EOG electrodes. The data were visually inspected to identify bad data spans (e.g., muscle artifacts) and noisy channels. These were interpolated using the default spherical spline interpolation. Independent component analysis (ICA) was performed on the continuous data using the default functions *ICA()* and *ica.fit()*, which were set to generate 35 components. Independent components corresponding to ocular artifacts were removed based on visual inspection. On average 2.11 ± 1.62 SD components were rejected per block. The final adjective was our critical word because it was the point at which the critical distinction between expressive (via emotional adjectives) and assertive (via physical adjectives) meanings became available to the participants. In addition, the final word represents the only position in the sentence where our full 2×2 design allowed us to isolate pragmatic meaning without confounds, and where interaction effects can be meaningfully interpreted as reflecting speech act processing rather than lower-level factors. Any differences detected at earlier positions in the sentence would instead likely reflect differences in person-perspective, verb conjugation, syntax or semantics, or even low-level visual differences across stimuli. Given that the critical word was the adjective in each sentence, the data were segmented between –100 and +800 ms

relative to the critical word onset (“ich bin froh”, “I am happy”) and baseline correction was applied using the -100 to 0 ms range as a reference (pre-word baselined data). Epochs where any channel exceeded a 200 μV peak-to-peak difference were rejected. First, data were averaged within subjects separately for all four experimental conditions. Next, the averages from the four conditions for each subject were grand-averaged across subjects.

To make sure that the results were not affected by our pre-word baseline, a separate analysis was conducted using a pre-sentence baseline instead of a pre-critical-word baseline. This produced epochs that were baseline-corrected just 200 ms prior to the onset of the first word of each trial (as opposed to the third critical word). This differs from the prior analysis where we chose a 100 ms baseline to avoid contamination from neural activity elicited by the preceding word, given the shorter ISI of 800 ms. In contrast, for the pre-sentence analysis, a longer 2000 ms ISI allowed the use of a 200 ms baseline without risk of overlapping from prior stimulus processing. For the pre-sentence baseline data, first a 1Hz high-pass filter was applied to the data to minimize the influence of electrode drift during these particularly long epochs. Then segmentation was performed from -2100 to $+800$ ms and baseline corrected using the -2100 to -1900 ms as a reference. All other processing steps were identical to the pre-word baselined data.

2.5. EEG analysis

To test our hypothesis that the communicative action would have an early effect on the ERPs, time windows of interest were selected in a data-driven, yet unbiased fashion, similar to previous studies on communicative action comprehension (Egorova et al., 2013; R. Tomasello et al., 2019, 2022). This was done by first examining the grand-average across all electrodes and, importantly, across all four experimental conditions collapsed. Peaks of average activity were detected, and time windows were chosen around these peaks using the

full width at half maximum (FWHM) approach. This resulted in four data-driven time windows of interest: (TW1) 76 – 100 ms, (TW2) 110 – 130 ms, (TW3) 143 – 193 ms and (TW4) 221 – 279 ms (Fig. 3). In addition, the N400 (300 – 450 ms) and early (450 – 600 ms) and late P600 (600 – 750 ms) components were examined too for reasons of comparability with previous studies, which have shown that these components can be affected by pragmatic factors (see e.g., Bambini et al., 2016; Noveck and Posada, 2003; Politzer-Ahles, 2020). Statistical analyses of the data were conducted on 36 frontal, central and posterior electrodes. These were further subdivided into three subsets according to the anterior-to-posterior gradient and another three subsets according to the left-to-right laterality. Crossing these two topographical factors therefore resulted in 9 local electrode clusters: anterior left (F7, F5, F3, FC5), anterior midline (F1, Fz, F2, FCz), anterior right (F4, F6, F8, FC6), central left (T7, C5, C3, CP5), central midline (C1, Cz, C2, CPz), central right (C4, C6, T8, CP6), posterior left (P7, P5, P3, PO7), posterior midline (P1, Pz, P2, POz) and posterior right (P4, P6, P8, PO8). This electrode selection is similar to that of Boux et al. (2021). The inclusion of Gradient and Laterality factors in the rmANOVA was motivated by prior work which demonstrated that different communicative actions are processed by different distributed networks of brain areas (Egorova et al., 2014; R. Tomasello et al., 2019, 2022; Boux et al., 2021), resulting in different EEG topographies. For these reasons, we considered it plausible that the Person $\times$ Adjective interaction would be specific to certain scalp locations rather than uniformly present across electrodes. We therefore also included the factors Gradient (anterior, central, posterior) and Laterality (left, midline right) to our analyses.

Statistical testing was conducted in Statistica 64 (version 12, StatSoft, Inc; Tulsa, OK, USA). For each time window, the normality of the data was not violated, as assessed by a Shapiro-Wilk test (all $p > 0.4$). This was the case both in the data with a pre-word baseline as well as for the data with the pre-sentence baseline. For each time window, a $2 \times 2 \times 3 \times 3$ repeated measures ANOVA was conducted with factors Person

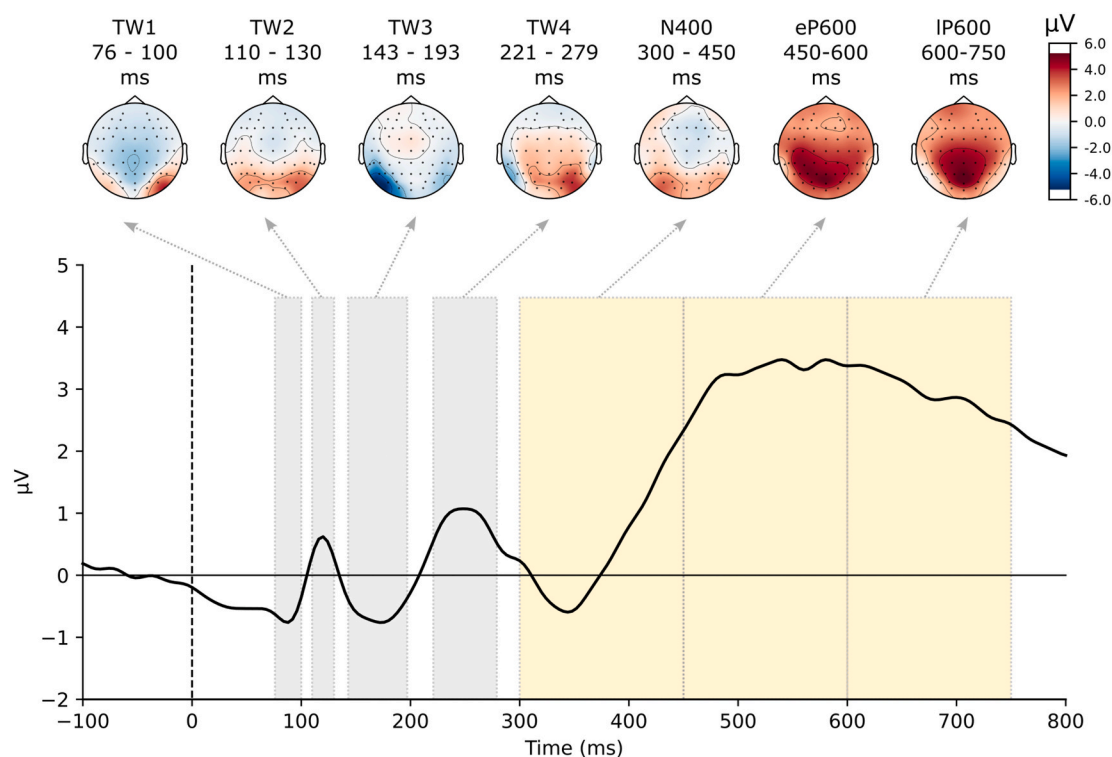


Fig. 3. Localizer ERP obtained by averaging potentials across all conditions, all electrodes and all subjects in the pre-word baselined data. Grey boxes indicate the four unbiased data-driven time windows defined around each peak using the FWHM method. The yellow boxes indicate the three a priori time windows taken from the literature, capturing the N400, early P600 (eP600) and late P600 (IP600). The upper banner shows the average topographical maps obtained for each of these time windows.

[1st/3rd] $\times$ Adjective [emotional/physical] $\times$ Gradient [anterior/central/posterior] $\times$ Laterality [left/midline/right]. Note that the assumption of independence of observations was not violated because the data from the two repetitions were collapsed, resulting in one observation for each subject and for each cell of the rmANOVA. Where this main analysis detected significant effects, we conducted two additional control

analyses on the pre-word baselined data. The first was to assess whether stimulus repetition affected our data. To this purpose, we ran a further rmANOVA with the addition of the factor Repetition [1st, 2nd]. The second was to assess generalizability across stimulus material. To this purpose, we aggregated the data by-item (i.e. F2 item analysis) and we ran a mixed ANOVA with Adjective as a between-item factor Adjective

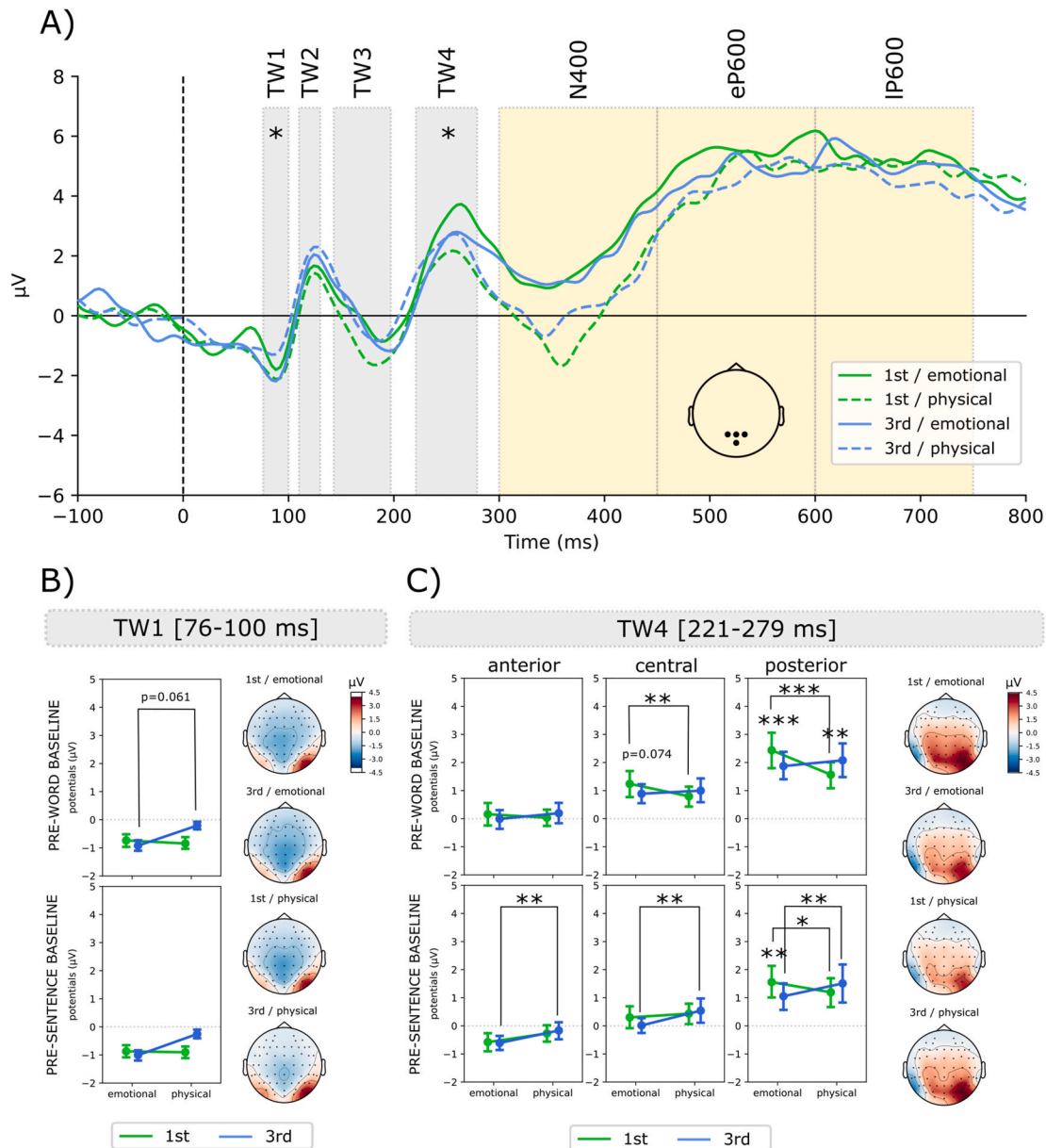


Fig. 4. Main results of the grand-average event-related brain potentials across all subjects ($n = 24$) in the relevant time windows (TW). Grey boxes indicate the four unbiased data-driven time windows defined around each peak using the FWHM method. The yellow boxes indicate the three a priori time windows taken from the literature, capturing the N400, early P600 (eP600) and late P600 (IP600). Asterisks indicate statistical significance according to the established standards (* indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$). (A) Grand-average event-related potentials at the medial subset (P1, Pz, P2, POz) of posterior electrodes, best illustrating the dynamics of the Person $\times$ Adjective interaction. Note that the effect was in fact significant on a larger set of posterior and central electrodes, see panel C of this figure. Data-driven time-windows are indicated as grey boxes whereas a priori time windows are indicated as yellow boxes. The asterisks indicate the time windows where a two-way or higher-level interaction involving the factors Person and Adjective was detected in the rmANOVA. (B) Post-hoc analysis of the potentials averaged in TW1. The dot plots indicate the averaged potentials within the time window both in the pre-word-baseline (upper panel) and pre-sentence baseline data (lower panel), with the error bars representing the SEM across subjects. The asterisks indicate statistical significance between conditions after Bonferroni correction for 4 comparisons. The topographical plots display the grand average potential distributions across the scalp in TW1 for each of the four conditions of the pre-word-baselined data. (C) Post-hoc analyses of the potentials averaged in TW4. The dot-plots indicate the averaged potentials within the time window at the anterior (F7, F5, F3, FC5, F1, Fz, F2, FCz, F4, F6, F8, FC6), central (T7, C5, C3, CP5, C1, Cz, C2, CPz, C4, C6, T8, CP6) and posterior (P7, P5, P3, PO7, P1, Pz, P2, POz, P4, P6, P8, PO8) electrode pools both in the pre-word-baseline (upper panel) and pre-sentence baseline data (lower panel), with error bars representing SEMs across subjects. The asterisks indicate statistical significance between conditions after Bonferroni correction for 12 comparisons. The topographical plots display the grand average potential distributions across the scalp in TW4 for each of the four conditions of the pre-word-baselined data.

and the within-item factors Person, Gradient and Laterality. For all ANOVAs, we report the partial eta squared (η_p^2) as a measure of the effect size. In addition, where the sphericity assumption was violated, the ϵ parameter as well as the Greenhouse-Geisser corrected p-values (p_{GG}) are reported (Greenhouse and Geisser, 1959). Post-hoc Fisher tests further corrected by Bonferroni correction were conducted to further characterize significant interactions. Bonferroni corrected p-values are reported (p_{Bonf}).

3. Results

3.1. Memory task

For each subject, the d' was calculated ($d' = z_{\text{hit rate}} - z_{\text{false alarm rate}}$) as a measure of their sensitivity in detecting the previously seen adjectives. Only one subject had a $d' = 0$ and was therefore excluded from further analyses on the basis of behavioral data (see Section 2.1). On average, the remaining subjects achieved a d' of 0.71 ± 0.27 SD in the unannounced recognition memory task, which was significantly different from 0, as evidenced by a one-sample t -test ($t(23) = 12.5$, $p < 0.001$). This indicates that they were attentive during the passive experimental task.

3.2. Brain potentials

For conciseness, in the main article text, we report only significant interactions involving the factors Person and Adjective, which, in the present design, are the only reliable indicators of a communicative action effect, as explained in the Introduction. We start with time windows where such a significant interaction was found. Other significant effects are reported in the [Supplementary Material](#).

Time Window 1 [76–100 ms]: The rmANOVA on the pre-word baselined data detected a significant Person $\times$ Adjective interaction ($F(1,23) = 4.56$, $p = 0.04$, $\eta_p^2 = 0.17$, see Fig. 4A). Post-hoc tests corrected for 4 comparisons (see Fig. 4B) indicated that this interaction was possibly driven by physical adjectives being more positive than the emotional adjectives in the 3rd person context ($p_{Bonf} = 0.061$) but not in 1st person sentences.

The pre-sentence epoch data, however, failed to confirm significance of the interaction between Person and Adjective ($F(1,23) = 1.51$, $p = 0.232$).

Time Window 4 [221–279 ms]: A significant Person $\times$ Adjective $\times$ Gradient interaction was detected ($F(2,46) = 4.84$, $p_{GG} = 0.034$, $\epsilon = 0.55$, $\eta_p^2 = 0.17$, see Fig. 4A). To further determine at which location along the anterior-posterior gradient the Person $\times$ Adjective interaction of interest was statistically significant, we conducted post-hoc tests corrected for 12 comparisons (see Fig. 4C). These showed that at posterior locations (P7, P5, P3, P1, Pz, P2, P4, P6, P8, PO7, POz, PO8), the emotional adjectives were associated with larger positivity when they appeared in a 1st person context than in a 3rd person context ($p_{Bonf} > 0.001$). Physical adjectives behaved the other way around so they were associated with less positive potentials when in the 1st than in the 3rd person context ($p_{Bonf} < 0.001$). Most importantly, the emotional adjectives' ERPs were significantly more positive-going than their physical counterparts when presented in the 1st person condition ($p_{Bonf} < 0.01$, see green curves in Fig. 3, panel A and C). As we will argue in the Discussion below, this interaction pattern is compatible with the interpretation that the larger response in the 1st person emotional sentences, which are the preferred tool for performing expressive communicative actions, reflects brain processes of understanding expressives. A similar effect was found at central electrodes (T7, C5, C3, CP5, C1, Cz, C2, CPz, C4, C6, T8, CP6), where the ERPs to emotional adjectives were more positive-going than those to physical ones in the 1st person condition ($p_{Bonf} < 0.01$) and where the emotional adjectives' potentials were marginally significantly more positive in the 1st than those of the 3rd person utterances ($p_{Bonf} = 0.074$).

When examining the pre-sentence-baseline data, a similar trend emerged. There was a significant Person $\times$ Adjective $\times$ Gradient interaction ($F(2,46) = 4.10$, $p_{GG} = 0.049$, $\epsilon = 0.57$, $\eta_p^2 = 0.15$). This was due to the fact that, at posterior locations, emotional adjectives in the 1st person context elicited greater positivity than in the 3rd person context ($p_{Bonf} = 0.002$) and greater than physical adjectives in the 1st person context ($p_{Bonf} = 0.048$). In addition, across all locations along the anterior-posterior gradient, ERPs to emotional adjectives were more negative-going than those for physical ones in the 3rd person condition (all $p_{Bonf} < 0.01$), whereas such a difference was not found for 1st person utterances.

Finally, our two additional control analyses replicated this general pattern. The first control analysis assessed whether stimulus repetition affected our main interaction (see full ANOVA table in Supplementary Materials, S3). While we did detect various significant effects of Repetition (as main effect and in interaction), the original Person $\times$ Adjective $\times$ Gradient interaction remained significant ($F(2,42) = 4.786$, $p_{GG} = 0.037$, $\eta_p^2 = 0.18$). In addition, the four-way Repetition $\times$ Person $\times$ Adjective $\times$ Gradient interaction was not significant, indicating that our main finding was not affected by the repetition of items ($p_{GG} = 0.481$). The second control analysis, that is, the by-item analysis, also replicated the pattern from the main by-subject analysis (see full ANOVA table in Supplementary Materials, S4). We detected the same Person $\times$ Adjective $\times$ Gradient interaction ($F(2,132) = 4.78$, $p_{GG} = 0.027$, $\eta_p^2 = 0.07$). The interaction was driven by the 1st/emotional stimuli being associated with larger positivity than the 3rd/emotional at posterior locations ($p_{Bonf} = 0.001$, closely mirroring the by-subject analysis).

Time Window 2 [110–130 ms] and Time Window 3 [143–197 ms]: No significant two-way or higher-level interactions involving both the factors Person and Adjective were detected, neither in the pre-word nor in the pre-sentence baselined analysis (see supplementary Materials, [Tables S1 and S2](#)).

A priori N400, P600 (early and late) Time Windows: No significant two-way or higher-level interactions involving both the factors Person and Adjective were detected, neither in the pre-word nor in the pre-sentence baselined analysis (see Supplementary Materials, [Table S4](#)).

4. Discussion

In this study, we asked at which point in time the neural indicators of the recognition of different expressive vs. assertive communicative actions first diverge, in order to draw conclusions about the temporal dynamics of pragmatic brain processes. Building upon previous work focusing mostly on requests (*directives*) and descriptive communicative actions (*assertives*), we here examined the brain correlates of language used to express internal states, in particular one's own emotions. These *expressive* actions, performed by applying utterances in the 1st person singular including an emotion adjective, such as “I am happy”, were compared with statements or assertive linguistic actions carried by the same sentence forms but containing matched external-state adjectives related to visually perceivable bodily properties, for example “I am blond”. Further conditions controlling for the semantic difference used the same sentences in the 3rd person singular (e.g., “he is happy/blond”), whereby both internal- and external-state expressions function as assertives. Therefore, by manipulating grammatical person and emotional vs. physical semantics orthogonally, this design enabled us to investigate a putative brain correlate of expressive communicative actions.

We found that neurophysiological signals of expressives and assertives differed already at ~ 250 ms after the onset of the sentence-final adjectives over a large posterior scalp area. This effect was manifest by a significant effect of our hypothesized interaction between Person and Adjective, driven by more positive-going event-related brain potentials for 1st person expressives as compared with 3rd person assertives, whereas a similar difference was absent for the semantically and

otherwise matched control conditions including physical assertions. In addition, we found another smaller and less robust effect very early-on, at ~85 ms, which, while being generally compatible with an effect of communicative action, remains difficult to interpret. Our results demonstrate specific signatures of expressive communicative action. They are consistent with and extend the previous findings that differences in the communicative action of utterances are reflected by distinct neural correlates. Our findings also show that pragmatic information can be processed quickly, within ca. 250 ms after the presentation onset of the stimulus critical for utterance understanding, in this case, the sentence-final adjective.

4.1. Neural dissociations of expressive vs. assertive communicative actions

Our main finding that the neurophysiological underpinnings for expressive vs. assertive communicative actions differ is based on the detection of significant interactions between Person and Adjective type in two separate time windows (221–279 ms and 76–100 ms post critical stimulus).

In the time window of between 221 and 279 ms after critical word onset, which still precedes the much-investigated ‘language potentials’ N400 and P600, the predicted interaction of interest was detected in both analyses conducted, with the data baselined either immediately prior to the critical word (pre-word baseline analysis) or prior to the beginning of the entire sentence (pre-sentence baseline analysis). The consistency of the results across analyses demonstrates the robustness of the effect. Specifically, the interaction effect, which was most pronounced at posterior scalp locations, indicated that sentences including emotional words elicited more positive-going event-related potentials in the 1st person than in the 3rd one, whereas this pattern was reversed (pre-word baseline analysis) or absent (pre-sentence baseline) for the same contrast in the physical sentences. Additional control analyses showed that this pattern was not affected by stimulus repetition and that it was also detectable when analyzing the data by-item rather than by-subject, underscoring the generalizability of this finding across items. Moreover, these neural differences are in line with the behavioral evidence from the rating study showing that our expressive and assertive communicative actions were indeed recognized as such.

It is important to clarify that the Person $\times$ Adjective interaction is the critical test in our design. The interaction isolates the pragmatic difference between expressive and assertive communicative actions. Examining only 1st person versus 3rd person sentences in the emotional adjective condition would introduce a confound, as these conditions differ not only in communicative function but also in grammatical perspective and verb conjugation. We controlled for these linguistic factors by including 1st and 3rd person sentences with physical (non-emotional) adjectives as baseline conditions. Similarly, comparing the neurophysiological responses for 1st person/emotional sentences with the 1st person/physical ones would also not be appropriate, as those have clear semantic differences. Therefore, we emphasize that the critical finding here is that there is a significant interaction showing that a modulation is happening that is non-additive, i.e., that goes above and beyond any mere effect of Adjective or Person alone. If this were not the case, then the utterances ending with emotional adjectives in the 1st and 3rd person would have elicited a pattern congruent with that seen for 1st and 3rd person sentences with physical adjectives. Instead, a significant interaction was observed across the four conditions (the four cells of the Person $\times$ Adjective design). This statistical result, regardless of the precise configuration of potentials in each condition, is in line with our hypothesis and provides evidence that the sentences preferably used for expressive communicative actions are processed uniquely against the control sentences inviting an interpretation as assertives. The significance of the Person $\times$ Adjective interaction (see Fig. 4, panels A and C) appears to be best explained by the communicative action of these linguistic structures, which are typical vehicles of communicative actions,

whereas all other conditions carried out assertive actions.

In the earlier time window (76–100 ms post critical stimulus), the interaction of interest was detected only when the pre-word baseline was applied to the data. In addition, the effect seemed to be driven by a scalp-wide lesser negativity for 3rd person physical sentences (e.g., “He is blond”), although the post-hoc tests remained marginally significant. If this significant interaction is also in line with our hypothesized communicative action effect, the fact that the very same interaction was not significant in the pre-sentence baselined data might be an indicator of a less reliable effect. It remains therefore difficult to provide a firm interpretation for this earlier time window.

A possibility worth some discussion is that our responses to expressive vs. assertive speech acts might have been linked in part to differences in arousal. Indeed, ratings provided on our stimulus material by an independent sample of subjects ($n = 12$) showed that emotional words embedded in 1st person sentences were found to be more arousing than the same emotional words embedded in 3rd person sentences (see Fig. 1) whereas the same effect was minimal for physical adjectives. However, we consider that, rather than a confound, it is an inherent property of emotion-related expressive communicative actions, due to their inherent link to expressing psychological states such as emotions (Searle, 1979). A sentence like “I am happy”, “sad” or “excited” is naturally more arousing than its physical counterpart *because* these sentence forms are normally used to express one's own emotions. We therefore suggest refraining from considering arousal as a confound but rather as an inherent property of expressives, which is manifest in the observed differences in ERPs.

4.2. Time-course of communicative action processing

As reviewed in the introduction, previous studies on the processing of expressives have mostly focused on attitude-expressives (Donahoo et al., 2022; Donahoo and Lai, 2020; Rohr and Abdel Rahman, 2018), that is, expressives that convey an attitude of the speaker towards certain things, people or events, rather than an emotional state. The processing of such attitude-expressives as opposed to more descriptive communicative actions was reflected in rather late phases of the ERPs, after 400 ms (Rohr and Abdel Rahman, 2018) or even 550 ms (Donahoo and Lai, 2020) post-critical stimulus. These results are in contrast with our present findings, which detect differences already at ~250 ms post critical stimulus, and not in the N400 or P600 time ranges. These differences can be explained in two ways. First, it is possible that these discrepancies reflect genuine differences in processing time course between emotion- and attitude-expressives, substantiating this distinction. Related to this is also the fact that the use of swear words (“damn”, “shit”, Donahoo et al., 2022; Donahoo and Lai, 2020) or of evaluative expressions directly addressed to the participants (e.g. “loser”; Rohr and Abdel Rahman, 2018) might have come with some additional dimensions of tabooeness or of social threat that were not present in the present emotion-expressives. However, why this would be reflected in later brain dissociations still remains open. A second possibility is that such previous studies might not have been able to detect earlier effects due to differences in design. In fact, all three mentioned studies use different critical words in their expressive vs. descriptive experimental conditions. Even if efforts to match the psycholinguistic properties of the stimuli were made, studies in which the same or near-identical verbal materials are used across conditions and where differences are controlled for by means of a factorial design (as in the present study) might allow more precise measures of the effects of interest and might be more sensitive to earlier (and typically smaller) effects. Finally, Donahoo et al. (2022) and Donahoo and Lai (2020) a-priori directed their analyses towards late rather than early time windows (for discussion, see Pulvermüller et al., 2009) and might, therefore, have missed earlier effects. Currently, it is not possible to distinguish between these possibilities. Nevertheless, our present study provides the first evidence that at least some types of expressives can be processed early, at ~250 ms

post-critical stimulus.

Previous studies have examined at which point in time the brain responses to perceived communicative actions, in particular assertives and directives, first diverge (Boux et al., 2021; Egorova et al., 2013, 2014, 2016; R. Tomasello et al., 2019, 2022; see also R. Tomasello, 2023; R. Tomasello et al., 2025). Interestingly, all of these studies gave evidence of early dissociations between communicative action types, typically at less than 300 ms and sometimes even <100ms after the relevant information was presented (Egorova et al., 2013, 2014; R. Tomasello et al., 2019, 2022). Our present study examining the neural signatures of expressive communicative actions vs. those of assertive communicative actions pointed to two different early timepoints of divergence between the two types of communicative action. The first one started already around 80 ms (see also Egorova et al., 2013, 2014; R. Tomasello et al., 2022), where the crucial interaction was significant, but less robust across baseline choices and difficult to interpret in terms of directionality. The second time window where significant effects emerged was around 250 ms, and was therefore similar in latency to those reported by previous studies (Egorova et al., 2013; Gisladdottir et al., 2015; R. Tomasello et al., 2019). Overall, the current results and previously reported ones consistently show that the brain can quickly distinguish between communicative actions, be they assertive, directive or expressive in nature. We note again that these ‘early’ responses occur before the onset of those ERP components most frequently examined in language research, including N400 and P600. This finding is in line with parallel models of language processing, which postulate that different aspects (e.g., phonological, semantic, syntactic and pragmatic) of a linguistic stimulus can be processed early (Marslen-Wilson, 1987; Pulvermüller, 2016; Pulvermüller et al., 2009; Strijkers et al., 2017; Strijkers and Costa, 2016). These models also propose that these various aspects of the linguistic stimulus can be processed early on and at similar latencies. Indeed, even if this was not the main objective of our study, we also detect an effect of Person (Person $\times$ Laterality interaction, see Table 1 of Supplementary Material) in the third time window (143–193 ms). This early effect can indeed be interpreted as a reflection of syntactic processing, although the person effect has an obvious confound linked to the different linguistic forms for expressing these syntactic properties (i.e., “ich bin” vs. “er ist”). Thus, overall, this study provides evidence for early indices of communicative action processing, ca. 250 ms after critical stimulus appearance (and possibly earlier). This result fits into a number of consistent results reported earlier showing phonological (Eulitz et al., 1995; Eulitz and Lahiri, 2004; Mesgarani et al., 2014; Näätänen et al., 1997), lexico-semantic (Hauk and Pulvermüller, 2004; Kiefer et al., 2008; Pulvermüller et al., 2009; Shtyrov et al., 2014) and morpho-syntactic (e.g., Friederici et al., 1993; Lucchese et al., 2017; Pulvermüller et al., 2008; Shtyrov et al., 2003) effects at early latencies post critical stimulus onset in comprehension, but also in production (Costa et al., 2009; Fairs et al., 2021; Feng et al., 2021; Miozzo et al., 2015; Rose et al., 2019; Rose and Abdel Rahman, 2017; Strijkers et al., 2010, 2011, 2017). In contrast, our results cannot be accounted for by classic models proposing serial processing of these various aspects of the linguistic stimulus, which would instead predict pragmatic information to be processed only at the end of a long processing cascade and certainly not prior to the ‘language components’ N400 and P600 (i.e., before 300ms; Friederici, 2002, 2012; Pickering and Garrod, 2004).

In the broader pragmatic literature, several pragmatic phenomena have been shown to be reflected mostly in late ERP components, especially the N400 and the P600 (Canal and Bambini, 2023). Such pragmatic phenomena include, among others, metonymy (Schumacher, 2013), scalar implicature (Noveck and Posada, 2003; see Politzer-Ahles, 2020 for a critical review) or metaphor (e.g., Bambini et al., 2016; Coulson and Van Petten, 2002). Only a few studies examining these phenomena detected earlier effects (see e.g., Domaneschi et al., 2018; Regel et al., 2011). In the present study, we did not detect any speech-act-related effect (i.e., no two-way interaction between Person

and Adjective) in the N400 or in P600 ranges. Yet we note that other studies which reported on early effects of communicative action also found additional late effects in the N400-P600 ranges too (Egorova et al., 2013; Gisladdottir et al., 2015; R. Tomasello et al., 2019). Overall, we would like to emphasize that linguistic pragmatic phenomena are varied in their nature and studies focusing specifically on the processing of communicative action type have consistently supported an early onset of processing. The fact that the processing of communicative action starts early on is not in contrast with the possibility that these and other pragmatic processes last throughout a longer time period or even start at a later point. Finally, a dissociation of brain responses at time t implies that at or before t , the underlying brain mechanisms make the related distinction; there is no implication about how long these processes last. As virtually any linguistic utterance is used by speakers to perform a communicative action, identifying the communicative action type is a very fundamental aspect of nearly every instance of language comprehension. Therefore, it might not be so surprising that the processing of information about communicative action starts earlier than that of other pragmatic or interpretation-related phenomena, such as irony or metaphor, which have been suggested to require additional inferences (Goodman and Frank, 2016; Grice, 1975; Noveck and Reboul, 2008).

Finally, here we examined the case of processing of communicative action at a timepoint where all information is available to the speaker to disambiguate the communicative action itself, that is, at the last word of the sentence. It is, however, quite possible that earlier words in the sentence already carry information that contributes to processing communicative action. For instance, in our assertive condition, it is possible that participants identified the assertive speech acts already at the appearance of the pronoun “er”/“he”. In the present study, we did not analyze brain potentials at such earlier words as these systematically come with a grammatical and visual perception confound stemming from simply comparing the 1st and 3rd person conditions (“he” vs. “I” and “is” vs. “am”). However, we do not exclude the possibility that these earlier linguistic cues could have contributed to pragmatic processing. Related to this, in the present study, we deliberately focused on the communicative action conveyed purely linguistically by presenting our stimuli in written form and in a minimalistic context provided by the instruction. While this approach was taken to maintain maximal experimental control, this still differs from natural written language, which is more contextualized (e.g., in prior discourse) and from spoken dialogue, which is additionally strongly enriched with a variety of non-linguistic cues. Particularly relevant here are signals conveyed by the speakers themselves in the case of face-to-face communication such as bodily signals which often occur prior to the corresponding utterance and can be predictive of pragmatic aspects thereof (Nota et al., 2021, 2023a, 2023b; Trujillo and Holler, 2021, 2024), but also prosodic ones (Hellbernd and Sammler, 2016, 2018; Villani et al., 2025). Such non-linguistic cues have also been found to be processed rapidly by the brain (R. Tomasello et al., 2019, 2022). Overall, our findings, combined with previous ones, are compatible with the view that the brain rapidly uses all cues, as soon as these are available (Pulvermüller, 2018), and uses them to probabilistically constrain the incoming message during comprehension (Altmann, 1998; Holler and Levinson, 2019; Tanenhaus and Trueswell, 1995). Future studies might attempt to approximate such naturalistic conversation while maintaining a certain degree of experimental control and explore the concurrent effect of multiple cues of different nature dynamically shaping the unfolding pragmatic processing.

4.3. Limitations and future perspectives

One might ask whether our present results might have been affected by possible confound factors. For instance, one could ask whether using different sets of words (emotional vs. physical) with a different visual appearance on screen, different semantics and differences in morphological complexity (emotional words were matched to physical ones for

various psycholinguistic variables, but not for morphological complexity, see Table 2) might have been factors influencing or even driving our results. Similarly, one could ask whether comparing 1st person sentences to 3rd person sentences might have had an effect which in part explains the present findings. Of course, it is well possible, and even consistent with pre-existing results (see, e.g., Section 4.2), that one or several of these factors generally affect the neural correlates of speech comprehension. However, these factors cannot on their own explain the interaction effect detected in the present study. In fact, our use of the two control conditions with physical adjectives (i.e., “I am blond”, “He is blond”) and the focus of our hypothesis on a Person $\times$ Adjective *interaction* rather than on a mere main effect of Person or of Adjective allows us to rule out these potential confounds. Indeed, the Person $\times$ Adjective interaction captures the neural differences above and beyond those due to grammatical Person (captured by the main effect of Person) and various potential formal, semantic and emotional-affective differences between adjective types (captured by a main effect of Adjective). Therefore, these other factors do not explain our main result. Note also that, due to these grammatical and lexico-semantic differences, it is not possible to interpret significant main effects of Adjective and Person in this study.

One further criticism could be that, given that the task was a passive task (i.e., no response was expected from the subject), one could ask whether participants were truly attentive to the stimulus material. We do find evidence that our subjects attended to the stimulus material. First, the electrophysiological dissociations that we found between the various experimental conditions would remain unexplained had the participant not attended to the stimulus material. In addition, the subjects' recognition memory was assessed after the EEG task. The d' in this memory test indicates that the subjects had a statistically significant, although modest, memory of the stimuli presented during the EEG task (mean $d' = 0.71$, $SD = 0.27$, see Section 3.1). The only subject with a $d' = 0$, demonstrating no sensitivity whatsoever to the material, was excluded. At first sight, it might seem surprising that, overall, subjects did not achieve a better memory performance. However, this is easily explainable. In fact, the memory task was unannounced, meaning that the subjects were not explicitly instructed to memorize the stimulus material. In addition, the adjectives used in the main task were semantically restricted as they were either emotional or physical adjectives. Furthermore, distractors in the memory test were taken from the same semantic categories as those used in the task and were therefore very semantically similar to the target words, making the detection of previously seen adjectives a difficult task. In light of these considerations, it becomes evident that the d' values achieved here cannot be compared to those achieved in typical online active tasks or to tasks where subjects are explicitly instructed to memorize the stimulus materials. Therefore, even our modest d' values can, combined with the differences in brain potentials, provide valid evidence that the subjects attended to the stimulus material during the EEG recording.

Finally, some criticism might be raised regarding our choice of stimuli. For instance, communicating other people's states of mind might include mechanisms of empathic resonance and, therefore, be partly also an expression of an internal state acquired by the speaker (Gallese, 2003). In other words, our control assertives in the “he is happy” form might have to a degree acted as “expressives” because they could have been interpreted as an expression of the empathic response of the person using the sentence. Related to this, while subjects were instructed that all sentence stimuli were being said by a close (male) friend, clearly indicating that they were not referring to the subjects themselves, we cannot exclude that our subjects might have engaged in self-reference to some degree when processing the stimuli and might have processed some sentences as if they were referring to themselves. In a first possibility, this might have happened preferentially for 1st person sentences, therefore correlating with the factor Person. In a second possibility, this might have happened randomly for all types of sentences, with no correlation with Person or Adjective type, especially in cases where the

adjective reflected some genuine physical or emotional characteristics of the participant. While both cases are possible, none of them should become manifest in the form of a Person $\times$ Adjective interaction and therefore should not be the cause of our detected interaction. Another possibility is that the stimuli might have been interpreted differently from a pragmatic perspective. For instance, we here instructed the subjects that the sentences were produced by a male friend, which might possibly have impacted pragmatic processing. Our pre-study ratings clearly indicate that our stimulus material was overwhelmingly interpreted as expected: the control conditions were interpreted as expressing the emotional state of the speaker only in a few cases (1st/physical: 6.1%, 3rd/physical: 0.5%, 3rd/emotional: 1.7%) whereas our expressive stimuli were recognized as expressives 97.8% of the time (see Material and Methods Section 2.1 and Fig. 1C). Nevertheless, the impact of speaker gender on communicative action interpretation could be more specifically investigated in future dedicated studies that explicitly manipulate this variable.

To sum up, our main finding that differences in potentials found at ~ 250 ms post-critical stimulus cannot be easily interpreted as resulting from missing attention or the use of different words or grammatical forms in the emotional and physical conditions. Therefore, the most plausible interpretation appears to be that these effects reflect the processing of expressive (vs. assertive) communicative actions. The association between these potentials and the arousal ratings for each item is also consistent with this view if the latter are seen as a consequence of the communicative action assigned to the constructions. Even if some criticism may be countered by the arguments discussed, there are additional questions left open by the present study. The most obvious one of these addresses the brain generators underlying the observed differences in surface ERP recordings. We found that the early ERP differences occurred at posterior recording sites, above parietal cortex, but this may only provide a hint towards the brain-internal sources of these differences. Methods with high spatial resolution, such as fMRI or distributed source analysis combined with high-density neurophysiological recordings, need to be applied to pinpoint the brain areas from which the documented ERP effects originated.

4.4. Conclusion

In the present study, we provided the first evidence that comprehending (emotional) expressive communicative actions used by a partner to share their own emotional states elicits neural ERP signatures which differ from those of comprehending assertive communicative actions. We find robust differences between the ERPs of these two communicative actions around 250 ms after the display of critical stimuli, the disambiguating sentence-final adjectives, at posterior scalp locations. Furthermore, there were additional neurophysiological differences at an even earlier timepoint (< 100 ms), which were, however, more difficult to interpret. These results converge with previous work reporting early effects of communicative action processing before the N400 and P600 components most frequently used in ERP experiments on language. Importantly, the present study extends these findings to a class of communicative actions—expressives, whose neurophysiological correlates are understudied—and therefore suggests that early processing of communicative action is a general feature of neuropragmatic processing in the human brain.

CRediT authorship contribution statement

Isabella P. Boux: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Milena R. Osterloh:** Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation. **Rosario Tomasello:** Writing – review & editing, Methodology. **Luigi Grisoni:** Writing – review & editing, Methodology. **Friedemann Pulvermüller:** Writing – review & editing, Writing –

original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.neuropsychologia.2026.109395>.

Data availability statement

Aggregated data and stimulus material are available on OSF at <https://doi.org/10.17605/OSF.IO/XTKRP>. The raw data that supports our study cannot be disclosed to the public because the consent form signed by our participants did not explicitly permit sharing individual subject data with anyone outside of the author team.

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